

Chinmay Omkar Bitne

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SUMMARY

Recent M.S. Information Technology graduate (AI Specialization) from Arizona State University, specializing in retrieval-augmented generation and agentic LLM systems. Built RAG pipelines that cut document retrieval latency 65% and lifted semantic retrieval precision from 62% to 89% during an AI Engineering internship at Virtusa. Deployed a full-stack RAG application, AnswerLeaf AI, to production on Render and Vercel.

EDUCATION

Arizona State University

May 2026

Masters, Information Technology

• GPA: 4.00

Sanjivani College of Engineering – Pune University

Jun 2024

Bachelor's, Information Technology

• GPA: 3.64

PROFESSIONAL EXPERIENCE

Virtusa Corporation

Apr 2024 - Jul 2024

AI Engineer Intern

Hyderabad, India

- Designed 2 RAG pipelines using LangChain for two internal use cases, reducing document retrieval latency by 65% and eliminating 40+ hours/month of manual lookups.
- Optimized transformer-based embedding models and prompt engineering pipelines, improving semantic retrieval precision from 62% to 89% for LLM-powered document automation workflows.
- Containerized two GenAI applications with Docker, set up CI/CD pipelines on Google Cloud Platform, and added evaluation-metric monitoring, which streamlined prototype delivery and ensured consistent performance during the internship

PROJECT EXPERIENCE

AnswerLeaf AI - Grounded Academic Research Workspace

- Built and deployed a full-stack workspace (Render backend, Vercel frontend) for evidence-grounded assignments with source ingestion, hybrid retrieval, source allow lists, authority-aware reranking, and clickable citations.
- Implemented the orchestration and API layers with LangGraph and FastAPI, backed by PostgreSQL/pgvector for vector and keyword retrieval.
- Added citation verification and documented a scoped five-case synthetic regression test with retrieval hit rate and valid-citation checks.

ASTRA AI - Voice-First Windows Assistant (In Development)

- Building a localhost-first Windows assistant with Gemini Live bidirectional audio in an Electron, React, TypeScript, Vite, and Express application.
- Connecting 50+ local tools for applications, files, games, media, system monitoring, reminders, and multi-step workflows.
- Integrating GCP for cloud-connected AI services while developing screen understanding, proactive alerts, Gmail/calendar briefings, and native system-tray integration.

NutriAssist AI - LLM-Powered Nutrition Assistant

- Fine-tuned Qwen LLM with LoRA and prompt engineering, achieving 92% accuracy on nutrition recommendations vs. 71% baseline, a 3× improvement in domain-specific task performance using transfer learning.
- Designed an agentic multimodal pipeline orchestrating model interactions with external tools (Gemini API for meal image detection) and persistent session memory.
- Containerized the assistant with Docker and version-controlled deployment updates for local testing and reproducibility.

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, R
- AI / Machine Learning:** Large Language Models (LLM), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, PyTorch, TensorFlow, Hugging Face, Agentic Workflow, Model Optimization, MLOps, Fine-tuning, Multi-agent Systems, Model Context Protocol (MCP), Vector Databases, PostgreSQL/pgvector, Embeddings, Prompt Engineering
- Cloud & Infrastructure:** AWS, Google Cloud Platform (GCP), Docker, CI/CD
- Frameworks & Tools:** Flask, Django, n8n, REST APIs, Git, AI Coding (Codex/Claude Code/Cursor)

ACHIEVEMENTS

- IEEE Paper Publication:** IEEE Publication: Author, “Generative AI Solution for Lane Departure, Pedestrian Detection, and Paving of Autonomous Vehicle” ICRTC 2023.
- Winner, Virtusa Jatayu S3 Hackathon:** Led ML pipeline development for predictive analytics using 5G sensor data.
- Best Project Award, Ural Federal University (URFU): Alternovation (Project)** - an AI-powered furniture visualization and cost-estimation prototype.

Certifications

- AWS Academy Graduate – Machine Learning Foundations: Covered core machine-learning concepts, AWS ML workflows, model development, evaluation, and cloud-based experimentation.
- AWS Academy Graduate – Machine Learning for Natural Language Processing: Explored NLP concepts, text preprocessing, embeddings, language models, and AWS-based approaches for building NLP applications.
- AWS Academy Graduate – Data Engineering: Covered data ingestion, transformation, storage, processing pipelines, and foundational AWS data-engineering services.